

AP PHYSICS 1: ALGEBRA-BASED • THE UNIVERSAL STUDY GUIDE

Targeting a 3, 4, or 5 — every concept tagged for your tier

Tier badges: [3] foundational · [4] adds for 4+ · [5] adds for 5

Targeting a 4 → study [3] AND [4] items. Targeting a 5 → study all three tiers.

PAGE 1 | MATH OF 3/4/5 • 7 UNITS • CONCEPTUAL > ARITHMETIC

About this exam. AP Physics 1 is the algebra-based intro covering kinematics, dynamics, energy, momentum, simple harmonic motion, torque + rotation, and (as of 2024-25 redesign) **fluids**. Calculator + reference table allowed throughout. **AP Physics 1 rewards CONCEPTUAL reasoning more than arithmetic** — a typical 5-pointer asks you to predict + JUSTIFY without numbers.

THE MATH OF YOUR TARGET

Target	MCQ (40q, 50%)	FRQ (4q, 50%)	Approx %	Tier
3 (PASS)	~17 / 40 (~43%)	~16/45 raw	~40-45%	[3]
4	~22 / 40 (~55%)	~24/45 raw	~52-58%	[3] + [4]
5	~28 / 40 (~70%)	~32/45 raw	~68-75%	All

Time: 80 min for 40 MCQ (~2:00 each), 100 min for 4 FRQs (~25 min each). FRQs include: **mathematical routines (1)**, **translation between representations (1)**, **lab/experimental design (1)**, **paragraph argument (1)**. Each weighted differently.

THE 7 UNITS — by exam weight (2024+ redesign)

#	Unit	Weight	Focus
1	Kinematics	10–15%	1-D + 2-D motion, projectiles. No air resistance.
2	Force & Translational Dynamics	18–23%	Largest unit. Newton's laws, friction, FBDs
3	Work, Energy, Power	18–23%	Work-energy theorem, conservation, springs (Hooke's)
4	Linear Momentum	10–15%	Impulse = Δp , conservation, collisions
5	Torque & Rotational Dynamics	10–15%	$\tau = rF$, rotational kinematics, rolling
6	Energy + Momentum of Rotating Systems	5–8%	I , KE_{rot} , angular momentum, conservation
7	Oscillations + Fluids	5–8%	SHM ($T = 2\pi\sqrt{m/k}$), pendulum, density, pressure, buoyancy

KEY INSIGHT [5] Units 2-3 = ~40% of the exam combined. Forces + energy dominate. Most FRQs combine these (incline + friction + speed at bottom = forces + energy).

KEY EQUATIONS — the must-knows [3]/[4]

KINEMATICS (constant acceleration):

$$v = v_0 + at \quad x = x_0 + v_0 t + \frac{1}{2}at^2 \quad v^2 = v_0^2 + 2a(x - x_0)$$

DYNAMICS: $F_{\text{net}} = ma$ $F_{\text{friction}} = \mu N$ $F_{\text{spring}} = -kx$

CIRCULAR: $a_c = v^2/r$ $F_c = mv^2/r$

GRAVITY: $F_g = Gm_1m_2/r^2$ $g = GM/R^2$ (at surface)

ENERGY: $KE = \frac{1}{2}mv^2$ $U_g = mgh$ $U_s = \frac{1}{2}kx^2$

$$W = F \cdot d \cdot \cos \theta \quad P = W/t = F \cdot v$$

$$E_{\text{mech}} = KE + U \quad (\text{conserved if no nonconservative work})$$

MOMENTUM: $p = mv$ $J = \Delta p = F_{\text{avg}} \cdot \Delta t$ $\Sigma p_{\text{before}} = \Sigma p_{\text{after}}$

ROTATION: $\tau = rF \cdot \sin \theta$ $L = I\omega$ $KE_{\text{rot}} = \frac{1}{2}I\omega^2$

SHM: $T = 2\pi\sqrt{m/k}$ $T_{\text{pendulum}} = 2\pi\sqrt{L/g}$

PAGE 2 | UNIT 1: KINEMATICS — 1-D + 2-D MOTION

Constant-acceleration kinematics [3]

THE 4 KEY EQUATIONS (use ONLY when a is constant):

$$v = v_0 + at$$

$$x = x_0 + v_0t + \frac{1}{2}at^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

$$x - x_0 = \frac{1}{2}(v + v_0)t$$

PICK THE EQUATION that uses what you know + asks for what you want.

Identify variables: v_0 , v , a , x , t – circle 3 knowns, identify the unknown.

Free fall + projectile motion [3]/[4]

- **Free fall [3]:** $a = -9.8 \text{ m/s}^2$ (or -10 for estimation). At PEAK, $v_y = 0$.
- **Projectile [3]/[4]:** independent x and y motion. x : $a_x = 0$, $v_x = \text{const}$. y : $a_y = -g$, varies.
- **Range maximized at 45° launch [4]**, given level ground + no air resistance.
- **[4] Time-of-flight** = $2 \cdot v_{0y} / g$ (level ground). Half time = peak.

Reading motion graphs [3]/[4]

Graph	How to read
Position vs time	Slope = velocity. Steeper = faster. Curve up = accelerating in +. Horizontal = at rest.
Velocity vs time	Slope = acceleration. AREA under curve = displacement.
Acceleration vs time	Area under curve = change in velocity.

KEY INSIGHT [4] Three-graph translation FRQ. Given one of position/velocity/acceleration vs t , sketch the other two. Use slope and area rules. Do NOT just "shift" the curve — derive each from the relationship.

Common kinematics traps [3]/[4]

- **[3] Confusing displacement with distance.** Displacement is vector (signed). Distance is total path length.
- **[3] At peak of toss:** $v = 0$ but $a \neq 0$ (still $-g$). Common MCQ trap.
- **[4] Average velocity** = displacement / time. NOT $(v_{\text{initial}} + v_{\text{final}})/2$ unless constant a .

PAGE 3 | UNIT 2: FORCES — NEWTON'S LAWS · FRICTION · FBDs

Newton's Laws [3]

N1: Object at rest stays at rest (or constant v) unless force acts. (Inertia)
 N2: $F_{\text{net}} = ma$ (net force is the SUM of all forces)
 N3: Action-reaction: equal magnitude, opposite direction, on DIFFERENT objects.

Free-Body Diagrams (FBDs) — required on every force FRQ [3]/[4]

- [3] Draw the object as a dot. Show ALL forces with arrows from the dot.
- [3] Label each force: F_g or W (gravity), F_N (normal), f (friction), F_T (tension), F_{app} (applied).
- [3] Choose axes — for incline problems, tilt axes so x = along slope, y = perpendicular to slope.
- [4] FBD point values are HUGE on FRQ. Often 2-3 points alone. Even if you can't solve, draw a complete FBD.

Friction [3]/[4]

STATIC friction: $f_s \leq \mu_s \cdot N$ (just enough to prevent motion; $\max = \mu_s \cdot N$)
 KINETIC friction: $f_k = \mu_k \cdot N$ (constant magnitude when sliding)
 Generally $\mu_s > \mu_k$.
 Friction OPPOSES relative motion (or attempted motion) — draw arrow opposite v .

Inclined planes [3]/[4]

Angle θ from horizontal:
 Component of gravity ALONG slope (down): $mg \sin \theta$
 Component PERPENDICULAR (into surface): $mg \cos \theta$
 Normal force: $N = mg \cos \theta$ Friction: $f_k = \mu_k \cdot mg \cos \theta$
 Net acceleration sliding DOWN with friction: $a = g(\sin \theta - \mu_k \cos \theta)$
 Sliding UP (decelerating with friction): $a = -g(\sin \theta + \mu_k \cos \theta)$

Tension, pulleys, Atwood [4]

- [4] Massless string + frictionless pulley = same tension throughout.
- [4] Atwood ($m_1 > m_2$): $a = (m_1 - m_2)g / (m_1 + m_2)$; $T = 2m_1m_2g / (m_1 + m_2)$.
- [4] Always pick a sign convention for each mass + stick with it. Many students lose points by mixing signs.

KEY INSIGHT [5] Connect FBD to N2 EXPLICITLY. Don't just write $F_{\text{net}} = ma$. Write: "Taking down as positive: $\Sigma F = mg \sin \theta - \mu_k mg \cos \theta = ma \rightarrow a = g(\sin \theta - \mu_k \cos \theta)$." Show each force's contribution.

PAGE 4 | UNIT 3: CIRCULAR MOTION · UNIVERSAL GRAVITATION

Uniform Circular Motion [3]/[4]

CENTRIPETAL ACCELERATION: $a_c = v^2/r$ (always toward CENTER)
 CENTRIPETAL FORCE: $F_c = mv^2/r$ (NET force, always toward center)

KEY POINT: F_c is NOT a NEW force – it's the NAME for the NET force that points toward the center.

PERIOD: $T = 2\pi r/v$ FREQUENCY: $f = 1/T$

What provides the centripetal force? [3]/[4]/[5]

Scenario	Centripetal force is...
Car turning on flat road [3]	Friction (static, between tires + road)
Ball on string circling [3]	Tension in string (horizontal: T; vertical: T + gravity at top)
Satellite orbiting Earth [4]	Gravity ($F_g = Gm_1m_2/r^2$)
Rollercoaster top of loop [4]/[5]	Normal + gravity (both DOWN). Min speed: $N = 0$, so $g = v^2/r \rightarrow v_{\min} = \sqrt{gr}$
Banked curve [5]	Horizontal component of normal force (and friction if applicable)

WATCH OUT [3] Do NOT call F_c "centrifugal". Centrifugal force is a fictitious force from a rotating frame. AP Physics 1 uses **inertial frames** — the only force is centripetal (toward center). Saying "centrifugal" loses you a point.

Universal Gravitation [3]/[4]

$$F_g = G m_1 m_2 / r^2 \quad G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2$$

Force on planet's surface: $F_g = mg$ where $g = GM/R^2$

CIRCULAR ORBIT – gravity provides centripetal force:
 $GMm/r^2 = mv^2/r \rightarrow v_{\text{orbit}} = \sqrt{GM/r}$

$$\text{KEPLER'S 3RD: } T^2 = (4\pi^2/GM) \cdot r^3 \rightarrow T^2 \propto r^3$$

KEY INSIGHT [4] Period of a satellite in orbit **DEPENDS ONLY ON RADIUS**. $T = 2\pi\sqrt{r^3/GM}$. Mass of satellite cancels. ISS at low orbit takes ~90 min; geostationary at higher $r = 24$ hours.

PAGE 5 | UNIT 4: WORK · ENERGY · POWER

Work [3]/[4]

Work done by constant force: $W = F \cdot d \cdot \cos \theta$
 θ = angle between force + displacement.

Work done by VARIABLE force: $W = \text{AREA under } F\text{-vs-}x \text{ graph.}$

Sign of work:

F same direction as motion ($\theta = 0^\circ$): $W > 0$ (positive work)
 F perpendicular to motion ($\theta = 90^\circ$): $W = 0$ (e.g., normal force, F_c)
 F opposes motion ($\theta = 180^\circ$): $W < 0$ (e.g., friction, decelerating)

Kinetic Energy + Work-Energy Theorem [3]/[4]

$$KE = \frac{1}{2}mv^2$$

$$\text{WORK-ENERGY THEOREM: } W_{\text{net}} = \Delta KE = \frac{1}{2}m(v_f^2 - v_i^2)$$

W_{NET} includes ALL forces' work (including friction!).
 Common use: object slides from rest down ramp + across floor with friction:
 $-\mu mg \cdot d$ (friction work) = $0 - \frac{1}{2}mv^2$ (ΔKE) $\rightarrow v = \sqrt{2\mu gd}$

Potential Energy + Conservation [3]/[4]

GRAVITATIONAL PE (near surface): $U_g = mgh$ (h = height above reference)
 SPRING PE: $U_s = \frac{1}{2}kx^2$ (x = displacement from equilibrium)

CONSERVATION OF MECHANICAL ENERGY (no nonconservative work):

$$KE_i + U_i = KE_f + U_f$$

$$\frac{1}{2}mv_i^2 + mgh_i = \frac{1}{2}mv_f^2 + mgh_f$$

WITH FRICTION: $KE_i + U_i = KE_f + U_f + |W_{\text{friction}}|$
 where $|W_{\text{friction}}|$ = energy LOST as heat = $\mu mg \cdot d$ (on horizontal)

Power [3]/[4]

Average power: $P_{\text{avg}} = W / \Delta t$
 Instantaneous: $P = F \cdot v \cdot \cos \theta$

Units: 1 watt = 1 J/s

KEY INSIGHT [5] Use **ENERGY** first when asked about **FINAL SPEED** — it's faster than kinematics + forces. Energy bypasses the path; only initial + final states matter. Especially powerful for incline + friction problems.

Springs + Hooke's Law [3]/[4]

Hooke's Law: $F_{\text{spring}} = -kx$ (negative because force opposes displacement)
 Spring PE: $U_s = \frac{1}{2}kx^2$

From rest at $x = A$ (compressed), released:
 Max KE at $x = 0$: $\frac{1}{2}mv_{\text{max}}^2 = \frac{1}{2}kA^2 \rightarrow v_{\text{max}} = A \cdot \sqrt{k/m}$
 SHM frequency: $\omega = \sqrt{k/m}$, $T = 2\pi\sqrt{m/k}$

PAGE 6 | UNIT 5: LINEAR MOMENTUM · IMPULSE · COLLISIONS

Momentum + Impulse [3]/[4]

$$p = mv \quad (\text{vector, same direction as } v)$$

$$\text{IMPULSE: } J = F \cdot \Delta t = \Delta p = m \cdot \Delta v$$

On a F-vs-t graph, impulse = AREA under curve.

$$\text{AVG FORCE: } F_{\text{avg}} = \Delta p / \Delta t$$

HOW TO REDUCE FORCE during collision: INCREASE Δt .

Examples: airbags, padded gymnastics mats, follow-through in catching.

Conservation of Momentum [3]/[4]

When net EXTERNAL force = 0 during interaction, total p is conserved. Most collisions + explosions: external forces (gravity, normal) cancel during brief interaction.

$$\Sigma p_{\text{before}} = \Sigma p_{\text{after}}$$

$$m_1 v_{1_i} + m_2 v_{2_i} = m_1 v_{1_f} + m_2 v_{2_f}$$

Conserved separately in x AND y. 2-D problems: write p_x and p_y equations.

Collision Types [3]/[4]

Type	Conserves	Identifying clue
Elastic [3]	p AND KE conserved	Bouncing balls, billiards (idealized)
Inelastic [3]	p only (KE lost to heat/sound/deformation)	Most real collisions
Perfectly inelastic [3]/[4]	p only; objects STICK together	Bullet embeds in block, train cars couple

PERFECTLY INELASTIC (1-D): $m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_f$
 Solve for v_f : a single unknown.

ELASTIC (1-D): Use BOTH p AND KE conservation. Two equations, two unknowns.
 Special: equal masses + one at rest → moving mass STOPS, other gets v.

KEY INSIGHT [4] On every collision FRQ ask: "Is KE conserved?" Elastic = yes. Anything else = no. Then: "What stays constant?" Always p (momentum). Sometimes KE.

PAGE 7 | UNITS 5-7: ROTATION · TORQUE · SHM · FLUIDS

Rotational kinematics [4]

LINEAR	↔	ANGULAR
x	↔	θ (radians!)
v	↔	$\omega = d\theta/dt$
a	↔	$\alpha = d\omega/dt$

Constant α equations (analog to constant a):

$$\omega = \omega_0 + \alpha t \quad \theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2 \quad \omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$$

Tangential relations: $v_t = r\omega$ $a_t = r\alpha$

Torque [3]/[4]

$$\tau = rF \cdot \sin \theta \quad (\text{vector; } \theta = \text{angle between } r \text{ and } F)$$

Maximum torque when force is PERPENDICULAR to lever arm ($\theta = 90^\circ$).

Zero torque when force is parallel to r .

ROTATIONAL EQUILIBRIUM: $\Sigma \tau = 0$ (no angular acceleration)

Useful for static problems: ladder against wall, beam balanced on pivot.

$$N_2 \text{ FOR ROTATION: } \Sigma \tau = I \cdot \alpha$$

Moment of inertia I [4]/[5]

Common I's (axis through center):

Hoop / thin ring:	$I = MR^2$
Solid disk/cylinder:	$I = \frac{1}{2}MR^2$
Solid sphere:	$I = \frac{2}{5}MR^2$
Hollow sphere:	$I = \frac{2}{3}MR^2$
Rod about CM:	$I = \frac{1}{12}ML^2$
Rod about end:	$I = \frac{1}{3}ML^2$

$$\text{ROTATIONAL KE: } KE_{\text{rot}} = \frac{1}{2} I \omega^2$$

$$\text{ANGULAR MOMENTUM: } L = I\omega$$

Angular momentum conservation [4]/[5]

[4] If $\Sigma \tau_{\text{external}} = 0$, L is conserved. Famous example: ice skater pulling arms in — I decreases $\rightarrow \omega$ increases $\rightarrow$ spin faster. L stays constant.

- [5] Rolling without slipping: $v_{CM} = r\omega$, so $KE_{\text{total}} = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}(M + I/r^2)v^2$.
- [5] Race down incline (no slip): solid sphere wins (smallest I), then disk, then ring (slowest).

Simple Harmonic Motion (SHM) [3]/[4]

$$\text{SPRING-MASS: } T = 2\pi \sqrt{m/k}$$

$$\text{PENDULUM (small } \theta): T = 2\pi \sqrt{L/g} \quad (\text{independent of mass!})$$

$$\text{Position: } x(t) = A \cos(\omega t) \quad \omega = 2\pi/T$$

$$\text{Max speed: } v_{\text{max}} = A\omega \quad (\text{at equilibrium})$$

$$\text{Max accel: } a_{\text{max}} = A\omega^2 \quad (\text{at extreme})$$

$$\text{Energy: } E_{\text{total}} = \frac{1}{2}kA^2 \quad (\text{constant}) = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 \quad (\text{always})$$

Fluids — new in 2024+ [4]/[5]

$$\text{DENSITY: } \rho = m/V \quad (\text{kg/m}^3)$$

$$\text{PRESSURE: } P = F/A \quad (\text{Pa} = \text{N/m}^2)$$

$$\text{Pressure in a fluid at depth } h: P = P_0 + \rho gh$$

$$\text{ARCHIMEDES' / BUOYANCY: } F_b = \rho_{\text{fluid}} \cdot V_{\text{displaced}} \cdot g$$

$$\text{Float condition: } F_b = mg \rightarrow \rho_{\text{object}} < \rho_{\text{fluid}}$$

$$\text{CONTINUITY (incompressible flow): } A_1 v_1 = A_2 v_2$$

$$\text{BERNOULLI (inviscid): } P + \frac{1}{2}\rho v^2 + \rho gh = \text{const along streamline}$$

KEY INSIGHT [4] Buoyancy intuition. Object floats: $\rho_{\text{object}} < \rho_{\text{fluid}}$. Fraction submerged = $\rho_{\text{object}} / \rho_{\text{fluid}}$ (an iceberg in seawater is ~89% submerged because $\rho_{\text{ice}} / \rho_{\text{seawater}} \approx 0.89$).

PAGE 8 | FRQ STRATEGY · 4-Q FORMAT · PARAGRAPH ARGUMENT

The 4 FRQs — predictable types [3]/[4]

Type	Recipe
Mathematical Routine	Multi-step calculation. Use formulas correctly. Show every step. Carry symbols, plug numbers at end.
Translation Between Representations	Given graph/table/words → produce graph/equation/words. E.g., "sketch the velocity graph corresponding to this position graph."
Lab/Experimental Design	Design experiment to measure X. Identify variables, list equipment, describe procedure, predict graph + how slope/intercept gives the answer.
Paragraph Argument (Qualitative-Quantitative Translation)	Write 4-6 sentence paragraph using physics principles to explain a phenomenon. Cite Newton's laws, conservation laws, etc. by NAME.

Paragraph Argument FRQ — the 5 sentence template [4]/[5]

S1 – STATE the law/principle by name.
 "By Newton's Second Law, the net force on an object equals its mass times accel."

S2 – IDENTIFY the relevant forces/quantities in the scenario.
 "In this scenario, gravity and normal force act on the object on the incline."

S3 – APPLY the law: write the relationship in this case.
 "The net force along the incline is $mg \cdot \sin \theta - \mu mg \cdot \cos \theta$."

S4 – DERIVE/COMPARE to get the conclusion.
 "Therefore the acceleration is $g(\sin \theta - \mu \cos \theta)$, independent of mass."

S5 – INTERPRET / connect to question asked.
 "This is why both objects reach the bottom in the same time despite different masses."

Lab/Experimental Design — must include [4]/[5]

- [4] **List all equipment** (specific: "meter stick," not "ruler"). Mention scale precision.
- [4] **Identify independent + dependent + controlled variables.**
- [4] **Procedure:** 4-6 numbered steps. Specific. "Drop the ball and time with a stopwatch" — say HOW many trials, what AVERAGE.
- [5] **Graph predict:** "Plot v^2 vs h . Slope = $2g$." Always pick variables that LINEARIZE the relationship.
- [5] **Source of error:** explicitly identify ONE realistic source and how to mitigate.

FRQ point-earning habits [3]/[4]/[5]

- [3] **Free-body diagram** on every force question. Often 1-3 points alone.
- [3] **Show fundamental equation BEFORE manipulating.** "From conservation of energy: $\frac{1}{2}mv^2 = mgh \rightarrow v = \sqrt{2gh}$."
- [4] **Carry symbols, plug numbers last.** Earns derivation points + reduces errors.
- [4] **Units on every numerical answer.** Missing units = lost point.
- [5] **Sanity-check answers at limit cases.** "If $\mu = 0$, this reduces to $g \cdot \sin \theta$ — correct for frictionless."

PAGE 9 | COMMON TRAPS · CONCEPTUAL CONFUSIONS · MCQ STRATEGY

Top conceptual traps [3]/[4]/[5]

- [3] At the peak of a toss: $v = 0$ but $a = -9.8 \text{ m/s}^2$. Many students put $a = 0$.
- [3] The "centripetal force" is NOT a separate force. It's the NET inward force, which IS gravity, friction, tension, etc.
- [3] Static friction can be LESS than $\mu_s N$. It's only equal to $\mu_s N$ when at maximum (about to slip).
- [4] Action-reaction pairs (N3) act on DIFFERENT objects. "Gravity pulls book down, table pushes book up" — these are NOT N3 pairs (both act on book). N3 pair to gravity-on-book = book-on-Earth (gravitational pull on Earth).
- [4] Normal force $\neq mg$ always. On an incline, $N = mg \cos \theta$. In an elevator accelerating up, $N = m(g+a)$. In free-fall, $N = 0$.
- [5] Conservation of energy includes all forms. Mechanical energy not conserved if friction/drag — but TOTAL energy is (some becomes thermal).
- [5] Pendulum period independent of mass + amplitude (for small θ). Common surprise. Period only depends on L and g .

MCQ Strategy [3]/[4]

- [3] Eliminate units-wrong answers. If the question asks for force, the answer must have units of N. Cuts options by 30-50%.
- [3] Limit-case test: pick the answer that behaves correctly at extreme values ($x \rightarrow 0$, $t \rightarrow \infty$, $\mu = 0$).
- [4] Multi-correct MCQ (2 of 5 correct). Read EACH option independently. Cross-check at the end.
- [5] When stuck, sketch a diagram + label. Most physics intuition unlocks visually.

Time strategy on FRQs [3]/[4]

100 min for 4 FRQs = ~25 min each.

Order to attack:

1. SKIM all 4 FRQs first (~3 min). Identify the easiest.
2. Do EASIEST FRQ first to build confidence + bank time.
3. Do hardest FRQ second (you have most time + energy).
4. Save the paragraph argument for when you have stable thinking.

If you're stuck: SKIP to next subpart. Each part is graded separately.
Even 'simple FBD + state law' on a hard problem earns partial credit.

PAGE 10 | SCORE LADDER · TIER CHECKLISTS · THE ONE THING

THE SCORE LADDER

For a 3 (Pass)	For a 4	For a 5
MCQ: 17-21 / 40	MCQ: 22-26 / 40	MCQ: 28-32 / 40
Constant-a kinematics, $F=ma$, KE/PE , $p = mv$ basics solid.	Add: friction + incline, energy with friction, perfectly inelastic collisions.	Mastery: rolling, angular momentum, SHM, fluids, paragraph argument.
Draw FBD on every force problem.	FBD + correct N_2 application along axes; energy + momentum problem combos.	Conceptual reasoning paragraphs citing laws by name; limit-case checks.
1-2 graphs interpreted.	All 3 motion graphs translated correctly.	Predict + sketch graphs; identify slopes + areas as physical quantities.

TIER-SPECIFIC TEST-DAY CHECKLISTS

For a 3 — focus only on these:

- ✓ MCQ: 17+ correct.
- ✓ FBD on every force FRQ subpart.
- ✓ Constant-acceleration kinematics (the 4 equations).
- ✓ $KE + U = \text{const}$ for energy without friction.
- ✓ Momentum conservation in collisions.
- ✓ At peak of toss: $v = 0$ but $a \neq 0$.

For a 4 — add these on top:

- ✓ MCQ: 22+ correct.
- ✓ Inclined plane with friction: $N = mg \cos \theta$, $f = \mu N$, $a = g(\sin \theta - \mu \cos \theta)$.
- ✓ Energy with friction: include $|W_{\text{friction}}|$ as energy lost.
- ✓ Perfectly inelastic vs elastic collision distinction.
- ✓ F_c provided by gravity / friction / tension / normal — IDENTIFY in each scenario.
- ✓ $T = 2\pi v(m/k)$ and $T = 2\pi v(L/g)$ memorized.
- ✓ Buoyancy: $F_b = \rho V g$; *floating fraction* = $\rho_{\text{obj}}/\rho_{\text{fluid}}$.

For a 5 — add these on top:

- ✓ MCQ: 28+ correct.
- ✓ Rolling without slipping: $KE_{\text{total}} = \frac{1}{2}(M + I/r^2)v^2$.
- ✓ Angular momentum conservation (skater pulling arms in).
- ✓ Paragraph argument: 5-sentence template with named law citation.
- ✓ Lab design: linearize relationship, identify slope as physical quantity.
- ✓ Limit-case checks: "if $\mu = 0$, this reduces to $g \sin \theta$."
- ✓ Distinguish action-reaction pairs (N_3) from balanced forces (N_1/N_2).

THE ONE THING THAT MATTERS MOST [3][4][5]: DRAW A FREE-BODY DIAGRAM AND CITE THE PRINCIPLE BY NAME. AP Physics 1 is decided on conceptual reasoning + setup, not arithmetic. "Use $F = ma$ " earns 0. "By Newton's Second Law applied to the block on the incline (with friction along the slope opposing motion), the net force is $mg \sin \theta - \mu mg \cos \theta$, so $a = g(\sin \theta - \mu \cos \theta)$ " earns full credit. **Always: FBD → cite law by name → write equation in this scenario → solve → interpret.**

If you do nothing else: master the 5-sentence paragraph argument template, draw FBDs religiously, and use ENERGY first when asked about final speeds.

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